

SurvivalGPU

survivalGPU: scalable survival analysis in R and Python

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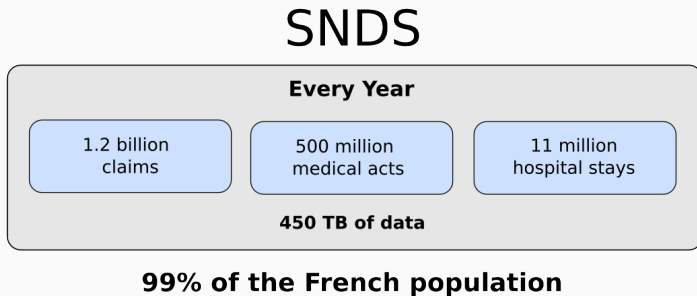
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Computational complexity in survival analysis

Claim databases: a gold mine for pharmaco-epidemiology¹



¹Health Data Hub, *Qu'est-ce que le SNDS?*, 2025.

The sources of complexity in the Cox model and its extensions

Number of lines

- Number of patients
- Time-dependent covariates

Number of covariates

- High-dimensional dataset
- Artificial covariates in the WCE model

Number of iterations

- Bootstraps
- Shapley values
- Cross-validation

Problem: computations can take a long time¹

2010 patients using hydroxychloroquine, 11 years of follow-up
WCE model (convolutive Cox model)
1000 bootstraps
3856 different events

CPU

5 days

¹Sabatier, Feydy, and Jannot, “Accelerating High-Dimensional Temporal Modelling Using Graphics Processing Units for Pharmacovigilance Signal Detection on Real-Life Data”, 2022.

GPU accelerated survival analysis



Computation of the convolutions



Maximization of the likelihood

Solution: GPU acceleration¹

2010 patients using hydroxychloroquine, 11 years of follow-up
WCE model (convolutive Cox model)
1000 bootstraps
3856 different events

CPU

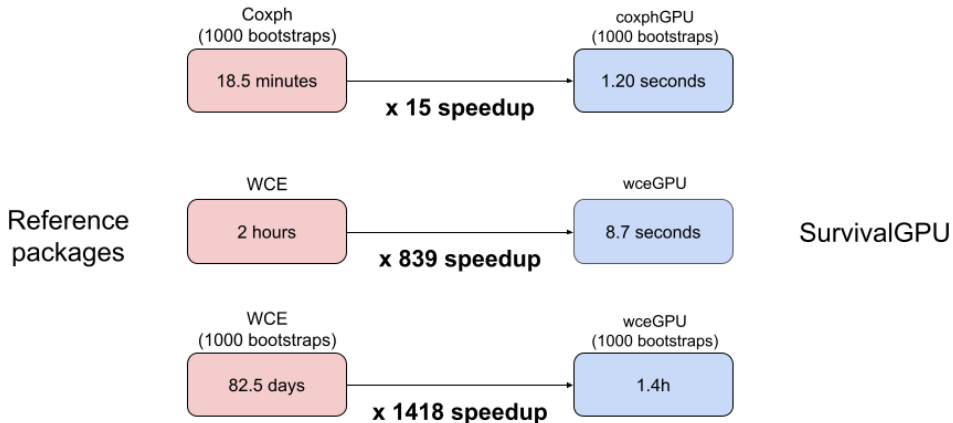
5 days

GPU

8 minutes

¹Sabatier, Feydy, and Jannot, “Accelerating High-Dimensional Temporal Modelling Using Graphics Processing Units for Pharmacovigilance Signal Detection on Real-Life Data”, 2022.

Benchmarks on simulated datasets of 100,000 patients



The survivalGPU package

Available models

- Coxph
- WCE

Following the convention of reference packages

- In R: R survival and the WCE package
- In Python: scikit-learn

The Cox model in R

```
library(survivalGPU)

model <- coxphGPU(Surv(Start, Stop, Event) ~ age + sex,
                  data = drugdata,
                  ties = "efron",
                  x = TRUE,
                  double_precision = FALSE)

model <- bootstrap(model,
                   R = 100,
                   patient_id = "Id",
                   data = drugdata,
                   batchsize = 25)
```

The Cox model in Python

```
from survivalgpu import CoxPHSurvivalAnalysis

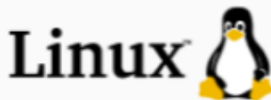
model = CoxPHSurvivalAnalysis(ties = "efron",
                              dtype = np.float32,
                              nbootstraps = 100,
                              batchsize = 25)

model.fit(covariates = drugdata[["age", "sex"]],
         start = drugdata["Start"],
         stop = drugdata["Stop"],
         event = drugdata["Event"],
         patient = drugdata["Id"])
```

Compatibility of the survivalGPU package



✓ Cox Model
✗ WCE Model



✓ Cox Model
✓ WCE Model



✓ Cox Model
✓ WCE Model

Windows Subsystem for Linux (WSL)

How to get access to an NVIDIA GPU?



How to find the package



Thank you for financing this work



References

-  Health Data Hub. **Qu'est-ce que le SNDS?** Documentation collaborative du SNDS. Accessed 07/09/2026. 2025. URL: <https://documentation-snds.health-data-hub.fr/snds/introduction/01-snds.html>.
-  Sabatier, Pierre, Jean Feydy, and Anne-Sophie Jannot. **“Accelerating High-Dimensional Temporal Modelling Using Graphics Processing Units for Pharmacovigilance Signal Detection on Real-Life Data”**. In: *Digital Health and Informatics Innovations for Sustainable Health Care Systems (MIE 2022)*. Vol. 294. Studies in Health Technology and Informatics. IOS Press, 2022, pp. 83–87. DOI: 10.3233/SHTI220401.